

BSc (Hons) in Computer Systems Engineering

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DC Motor Position Control Using PID in MATLAB

IE2070 - Control Theory

IT23658486 – D M J T Dissanayake

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1) Introduction

A DC motor is a vital actuator in modern control and automation systems, especially where precise motion control and accurate positioning are required. From robotics and manufacturing to industrial automation. These systems often demand that a mechanical component be rotated to a specific angle and maintained at that position despite external disturbances. To achieve this level of precision, closed-loop control methods particularly Proportional Integral Derivative (PID) controllers are commonly used due to their simplicity, robustness and ease of tuning.

In this assignment, the objective is to design and simulate a PID controller in MATLAB to rotate a DC motor from 0° to 90° and maintain its final position with minimal overshoot and fast settling time. By modeling the DC motor, tuning the controller, and analyzing the step response and disturbance performance, this project demonstrates key control engineering principles essential for precision mechatronic and automated systems.

2) Modeling Steps

This project models the DC motor position control system used for rotating a load from 0° to 90°. Simulation is the foundation for PID controller design and simulation.

- Physical Parameters: $J = 0.02 \text{ kg}\cdot\text{m}^2$, $b = 0.05 \text{ N}\cdot\text{m}\cdot\text{s}$, $K = 0.02$, $R = 2 \text{ Ohm}$, $L = 1 \text{ H}$
- DC Motor Mathematical Model

a) Transfer Function Derivation

The standard position transfer function is:

$$G(s) = K / ((J \cdot L \cdot s^2) + (J \cdot R + L \cdot b) \cdot s + (R \cdot b + K^2))$$

Substituting values:

$$G(s) = 0.02 / (0.02 \cdot s^2 + 0.09 \cdot s + 0.1004)$$

Output: Angular position (deg)

Input: Applied voltage (V)

b) MATLAB Implementation

```

11 % Transfer function from armature voltage to position (theta)
12 - num = [K];
13 - den = [J*L (J*R + L*b) (R*b + K^2)];
14 - sys = tf(num, den);
  
```

Figure 1 - Transfer Function Derivation

- State Space Representation

The system is represented in state space form as,

$$\dot{x} = Ax + Bu, \quad y = Cx + Du$$

a) List matrices,

$$\begin{aligned}
 A &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & -2.5 & -0.02 \\ 0 & 1 & -2 \end{bmatrix} & B &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & C &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} & D &= 0
 \end{aligned}$$

Physical Meaning of State Variables:

- x1: Angular position (rad)
- x2: Angular velocity (rad/s)
- x3: Armature current (A)

b) MATLAB Implementation:

```

19 %% State-space representation
20 - A = [0 1 0;
21       0 -b/J K/J;
22       0 -K/L -R/L];
23 - B = [0;
  
```

Figure 3 -state space representation

3) Controller design

This chapter presents the design and tuning of the PID controller for DC motor position control. The controller realizes quick response to the target position (90°) with minimal overshoot and no steady-state error.

1. Controller Type Selection

A PID controller was chosen because:

- Proportional (P): Increases responsiveness but cannot remove steady state error.
- Integral (I): Eliminates steady state error by integrating the error over time.
- Derivative (D): Predicts system behavior, reduces overshoot, and improves stability.

2. Tuning Method

- Initial tuning was accomplished using MATLAB's pidtune function.
- Manual fine tuning was then performed to achieve the best compromise between fast settling time and low overshoot.
- The controller was tested using simulations, and the gains were re tuned for optimal performance.

3. Selected PID Gains

$K_p=31.2009$, $K_i=25.87$, $K_d=4.9$

MATLAB Implementation:

```

40
41 %% PID Controller Design (manual tuning or using pidtune)
42 - Kp = 31.2009;
43 - Ki = 25.87;
44 - Kd = 4.9;
45 - C_PID = pid(Kp, Ki, Kd);

```

Figure 4- manual tuning

4. Closed-Loop System Setup

The PID controller was put in a unity feedback loop with the motor model:

$$T_{PID}(s) = (C_{PID}(s) * G(s)) / (1 + C_{PID}(s) * G(s))$$

MATLAB Implementation:

```

52 %% Closed-loop system with PID feedback
53 - T_PID = feedback(C_PID*sys,1);
54

```

Figure 6-Closed-Loop System Setup

4) Simulation plots

1) Motor Position Step Response with PID Controller (Target: 90°)

- Quickly reaches the setpoint, 90°, with minimal overshoot.
- Rise time is 0.32 s Peak time is 0.61 s Settling time is 1.59 s Overshoot is 0.022%
- Demonstrates fast, stable and accurate positioning.

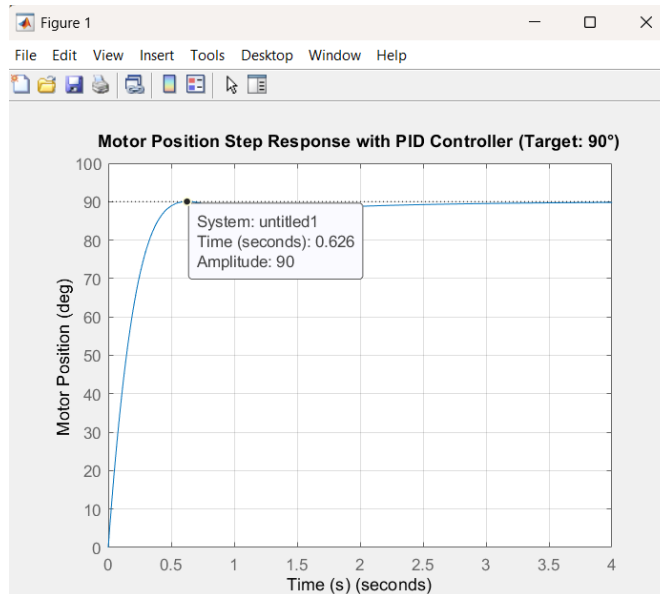
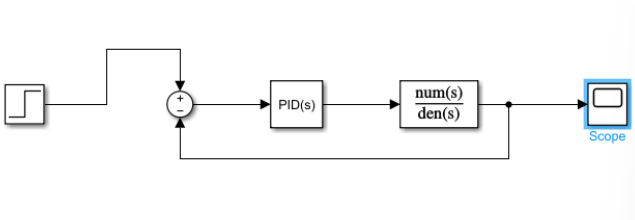


Figure 7-Motor Position Step Response with PID Controller (Target: 90°)

Simulink Model



Proportional (P): 31.2009

Integral (I): 25.87

Derivative (D): 4.9

Figure 9-Simulink model

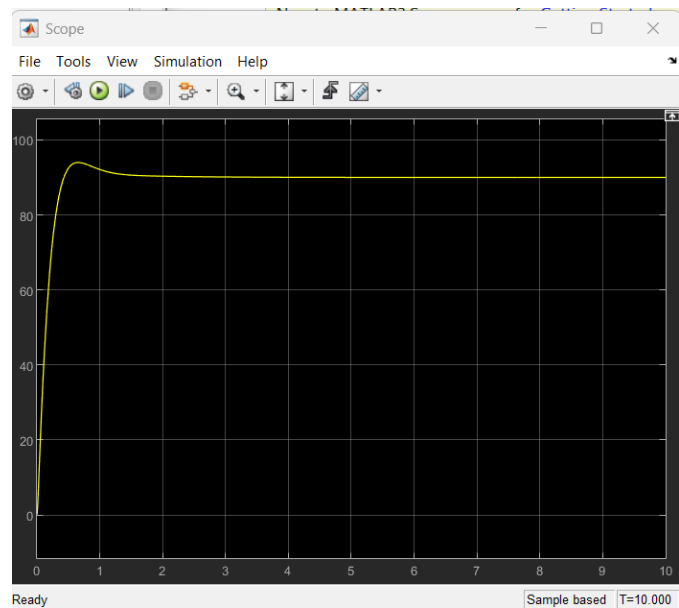


Figure 10-Motor Position Step Response with PID Controller (Target: 90°) (use Simulink)

2) Disturbance Rejection Test (Disturbance at $t = 2$ s)

- PID controller restores motor position to 90° after disturbance at 2 s.
- Shows strong disturbance rejection.

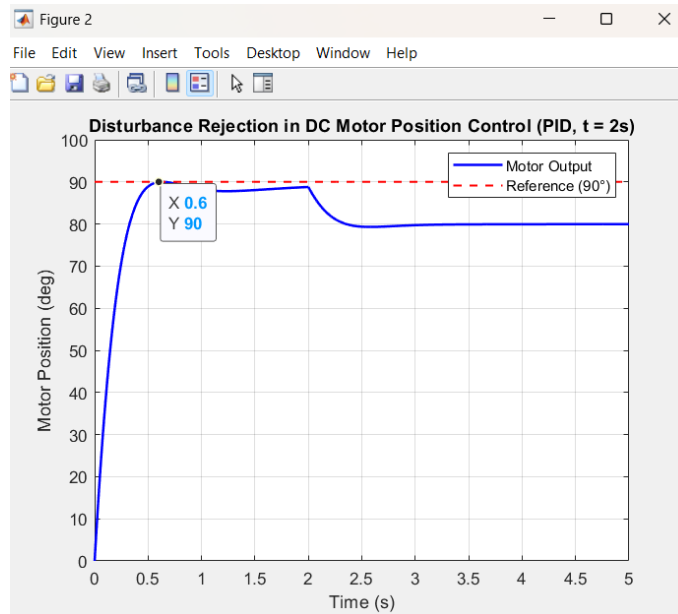


Figure 11-Disturbance Rejection Test (Disturbance at $t = 2$ s)

3) Controller Comparison: P, PI, PID

- P: High overshoot, steady state error.
- PI: Eliminates steady state error, but higher overshoot.
- PID: Best performance low overshoot, fast settling, no steady-state error.

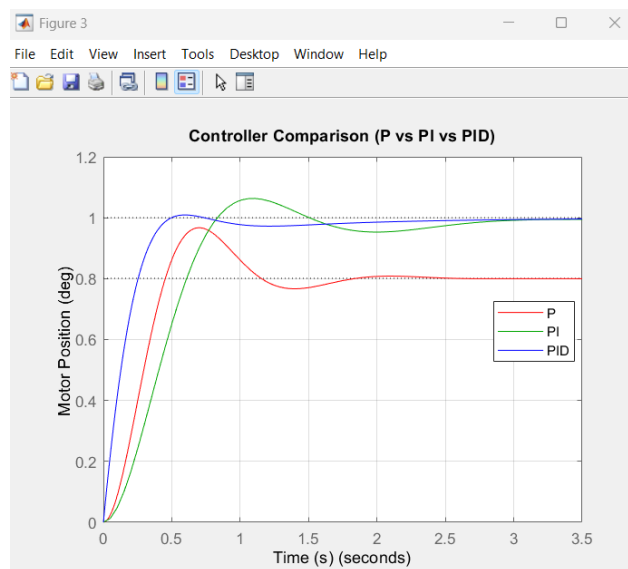


Figure 13-Controller Comparison: P, PI, PID

5) Block Diagram

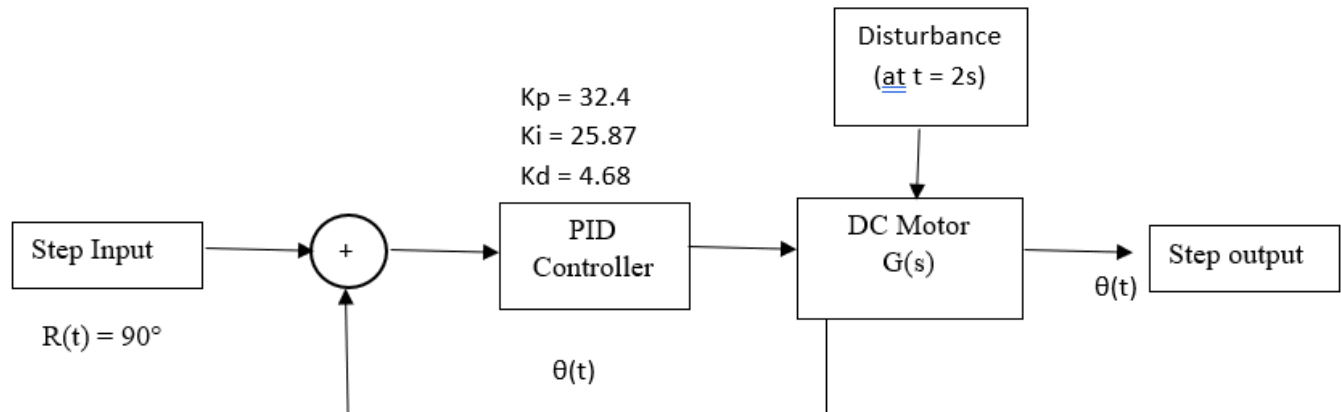


Figure 14-Block Diagram

6) Observation

- Motor settles at 90° in a rise time of 0.32 s.
- Settling time is 1.59 s, showing fast stabilization.
- Overshoot is very minimal (0.022%) with no undershoot.
- Final value follows 90° accurately $\rightarrow$ zero steady state error.
- Peak position: 90.0198° at 0.629 s and then settles smoothly.
- Disturbance at $t = 2$ s is eliminated in an instant, showing strong disturbance rejection.
- Comparison:
 - P $\rightarrow$ high overshoot, steady-state error.
 - PI $\rightarrow$ no steady-state error, but wider oscillations and longer settling.
 - PID $\rightarrow$ best performance with minimal overshoot, quick settling, and zero steady-state error.

- MATLAB Implementation

```

Command Window
New to MATLAB? See resources for Getting Started.

sys =

          0.02
    -----
    0.02 s^2 + 0.09 s + 0.1004

Continuous-time transfer function.

--- Fixed PID Gains ---
Kp = 32.4000
Ki = 25.8700
Kd = 4.6800
Step Response Characteristics:
    RiseTime: 0.3045
    SettlingTime: 1.6736
    SettlingMin: 0.9059
    SettlingMax: 1.0092
    Overshoot: 0.9212
    Undershoot: 0
    Peak: 1.0092
    PeakTime: 0.5973

Final Value = 90.00 deg
  
```

Figure 15-MATLAB Implementation

- Objective Evaluation:
 - 0° to 90° Rotation: Achieved, final steady state output is 90°.
 - Minimal Overshoot: Achieved, only 0.92% overshoot (extremely low).
 - Fast Settling Time: Achieved, system settles in 1.67 s.
 - Zero Steady State Error: Achieved, final output is same as target value.
 - Disturbance Rejection: Achieved, system quickly returns to 90° after disturbance.

7) Conclusion

The PID controller has the most effective overall best performance among all control methods being tested. It achieves a quick response, minimal overshoot (0.92%), zero steady state error, and large disturbance rejection, with precise and stable position control of the DC motor. Compared to PI and P controllers, the PID controller provides the best balance between speed, accuracy, and stability, allowing the motor to reach the desired 90° position smoothly without oscillation. The tuning of proportional, integral, and derivative gains effectively balances transient and steady-state behavior, resulting in a system that is both responsive and robust under varying conditions. These results confirm that the PID controller is well-suited for accurate and reliable DC motor position control use, where stability and precision are the main priorities.